Initial Value Problems and Euler's Method

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1 Introduction

A common type of problem that is encountered in the study of differential equations is the Initial Value Problem (IVP). IVPs can also be one of the more simple types of differential equations to solve, as the initial condition (e.g. $y(0) = 1$) can be used to perform substituiton and isolate unknowns. However, many differential equations, and even some IVPs, are not as straightforward in solving. This is where Euler's Method comes into the discussion.

Euler's Method is used when no solution can be found to a differential equation by hand. It produces an approximation by computing the slope at a point of the function (using the derivative) and moving a step size h. It then calculates a new slope and moves again. Using this method, error increases as more approximations are performed, as the approximation moving along h doesn't follow the true curve exactly. This tendency is shown in Figure 1, which demonstrates how the straight-line Euler's Method approximations deviate from a true function's line (in the figure, e^{-x}). The error of Euler's Method becomes much worse as the steepness of the curve rapidly changes. A further discussion of the accuuracy of Euler's Method will be explored below. Euler's Method is defined as follows:

$$w_0 = y_0, \qquad w_{i+1} = w_i + hf(t_i, w_i)$$

Where w_0 and y_0 indicate the initial value, w_i is the numerical approximation of the solution at the current time and w_{i+1} is the numerical approximation of the solution at the next time step t_{i+1}. h is the step size, and t_i is the time or independent variable at the i-th step. $f(t_i, w_i)$ is the differential equation function evaluated at the current time t_i and current approximation w_i.

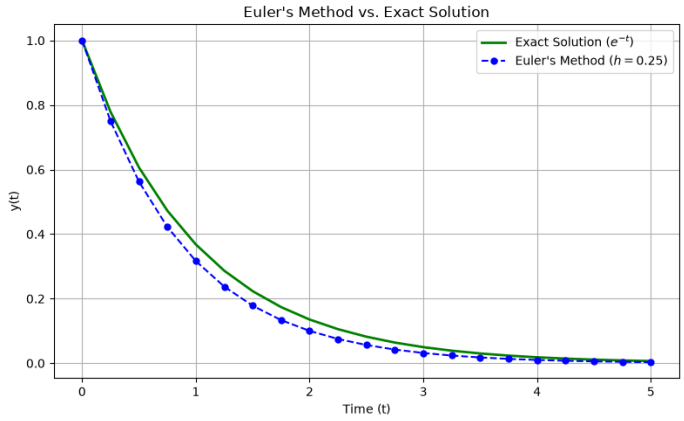


Figure 1: Error Tendencies of Euler's Method

2 The Problem

The example problem used in this discussion is

$$y' = ty + t^3, \qquad y(0) = 1$$

with initial condition specified above. Now, it is possible to find the solution to this equation by using differential equation solving skills. Thus, the exact solution is

$$y(t) = 3e^{\frac{t^2}{2}} - t^2 - 2$$

We will be using this to compute the accuracy of Euler's method and compare how accuracy increases as h decreases. Computing the error at several mesh resolutions (h values), the values in Table 1 are obtained. As demonstrated,

Table 1: Convergence Performance of Euler's Method

Step Size (h)	Euler Approx (t=2)	Exact Value (t=2)	Absolute Error
0.2	10.205253	16.167168	5.961915
0.1	12.646560	16.167168	3.520609
0.05	14.233190	16.167168	1.933978
0.02	15.345157	16.167168	0.822012
0.01	15.747429	16.167168	0.419740
0.005	15.955047	16.167168	0.212121

smaller step sizes of h result in improved accuracy. This is further seen in Figure 2, which shows how closely Euler's approximation gets to the true function at different values of h.

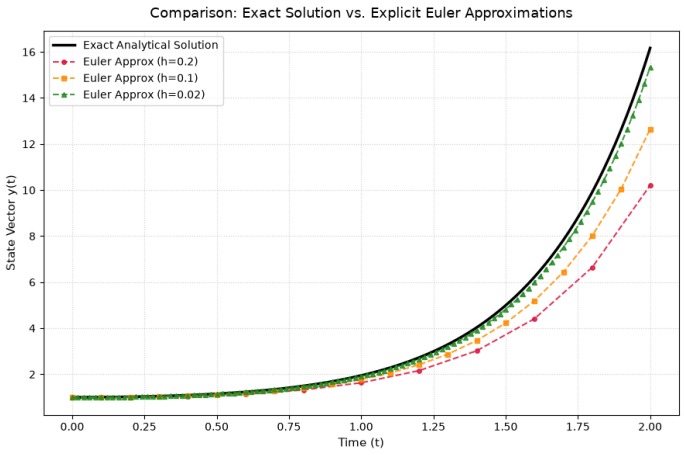


Figure 2: Euler's Method Compared to Exact Function

3 Error Considerations

Euler's Method is useful, but to use it effectively we must be familiar with the errors it is prone to and how to mitigate them. Locally, Euler's Method has a truncation error of order $\mathcal{O}(h^2)$. This is because, as h decreases the approximation rapidly improves. Globally, however, Euler's Method is of order $\mathcal{O}(h)$. This results from the relationship between increasing accuracy as $\lim_{h\to 0}$, and decreasing accuracy as n (number of steps) increases. This reflects the inverse relationship between h and n, and is further exemplified by Figures 3 & 4. Figure 5 shows the global, first-order convergence of Euler's Method and verifies the accuracy of our process of evaluation.

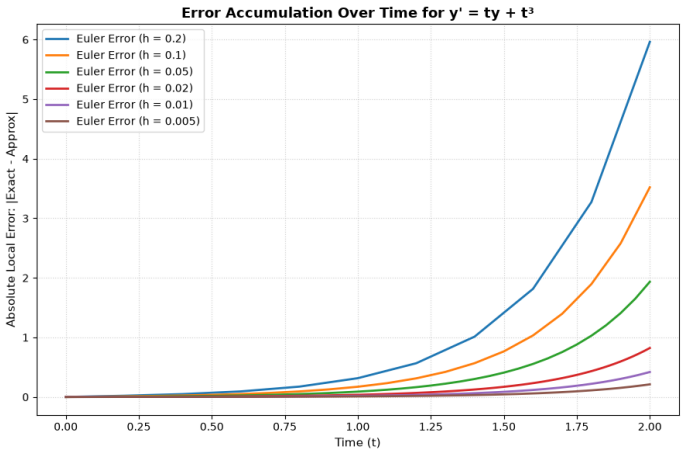


Figure 3: Error vs Time at different values of h

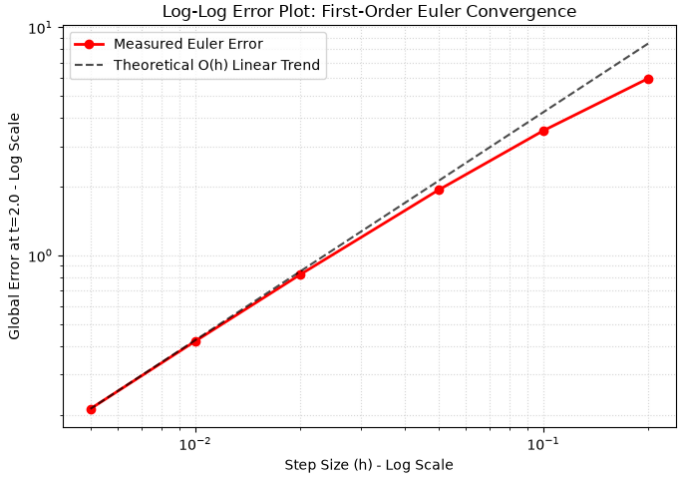


Figure 4: Convergence of Euler's Method: Global First-Order

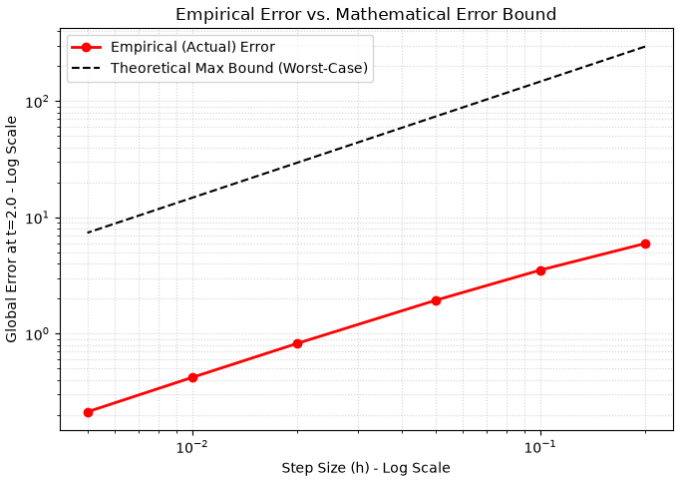


Figure 5: Empirical Error Bound

4 Conclusion

In this paper, we analyzed the effectiveness of Euler's Method and applied it to an example problem. The relationship between error and h was evaluated, and the local vs global convergence patterns of Euler's Method were considered. Euler's Method is overall, a consistent approximation method for functions, but it would still be desirable to find a method that follows the true curve better while not being overly complex to implement using computation.